

POLYP CLEAN PATHWAY

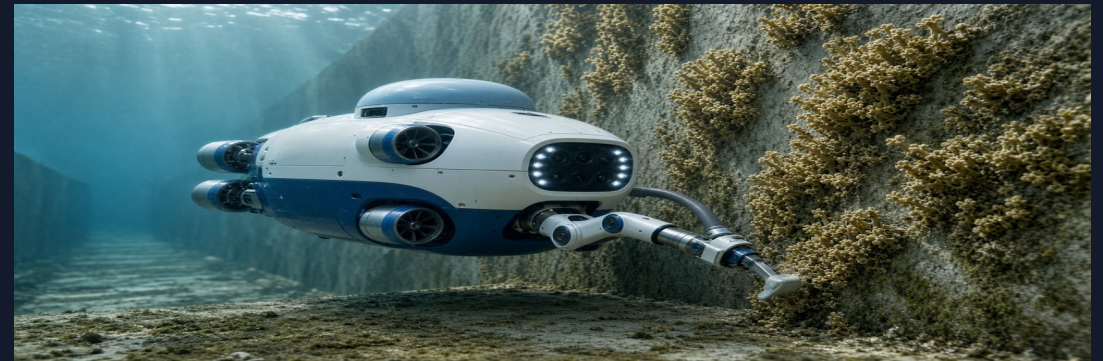
Block the inlet. Clean the channel. Protect Barakah.

A dual pathway for seasonal jellyfish blooms: a marine mesh at the seaward mouth of the cooling-water intake, plus an autonomous robot that removes polyps from the internal channel walls. The robot remains the core idea. The mesh is the front door.



LAYER 1 — INLET MESH

Self-cleaning exclusion grid at the labelled Inlet. Adult jellyfish stay in the Gulf. Cooling water still flows.



LAYER 2 — CHANNEL ROBOT

PolypClean AUV patrols walls, breakwater face and forebay. Gentle scrape + suction so polyps cannot seed a bloom inside the waterway.

Designed for a real intake — Barakah

About PolypClean Pathway

We redesigned the concept around the cooling-water intake of Barakah Nuclear Energy Plant (Al Dhafra, Abu Dhabi). The long sheltered channel is both the risk and the solution space.

If jellyfish never enter the mouth, and polyps never colonise the walls, the plant does not fight a bloom at the screens.

Scope stays inside the ENEC brief: we do not redesign plant systems or nuclear safety systems. We add a marine-side pathway that starts after the existing early-warning notification.

Focus: marine robotics · smart barrier & deflection · environmental tech · sustainability.

Team PolypClean

Robotics lead

AUV, thrusters, channel nav

Barrier systems

Mesh, self-clean, hydraulics

AI / vision

Polyp detection on walls

Marine ecology

Impact and coexistence

Systems

Deployment & approvals path

After the warning, the inlet is still open

ENEC already detects blooms offshore. The remaining gap is management once that notification is issued — without extra diver hours, and without harming marine life. Cooling-water intake blockage is a documented global NPP risk.



Barakah intake — seaward mouth labelled Inlet. This is the chokepoint.

01 Reactive only

Adults are handled after they arrive. Polyps on hard walls keep seeding the next wave.

02 Manual and risky

Diver and net work in a live intake is slow, weather-limited, and unsafe.

03 The channel itself

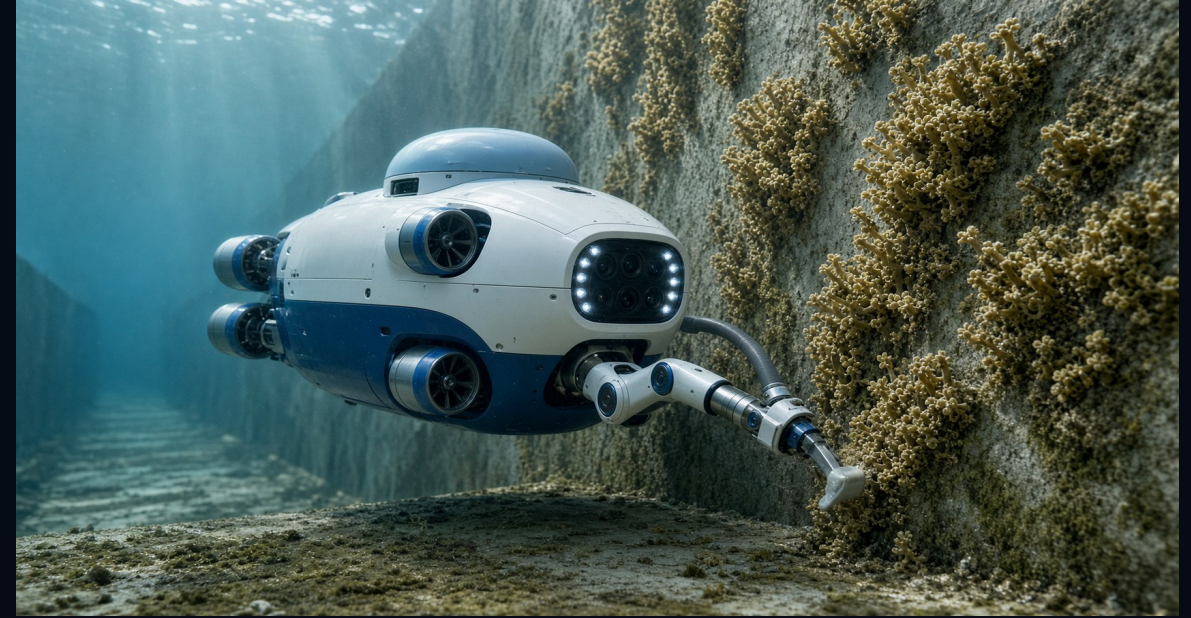
A long sheltered waterway (breakwater + walls + forebay) is ideal polyp habitat if left uncleaned.

Main idea: the robot. Added: the pathway.



LAYER 1 Inlet mesh (smart barrier)

Installed at the seaward inlet between breakwater arms. Excludes adult jellyfish while allowing through-flow. Modular panels + self-cleaning (brushes / backwash) so biofouling does not become a new blockage. Can be maintained without redesigning plant systems.



LAYER 2 PolypClean AUV (robotics)

Works inside the channel after the mesh. AI vision finds polyp colonies on walls, rocks and concrete. Robotic arm with gentle scraper / micro-cutter and suction collects material so it cannot reattach. Even if larvae pass the mesh, they cannot grow a nursery on the walls.

Two jellyfish origins, two defenses — not a single point fix

Why Mesh + Robot

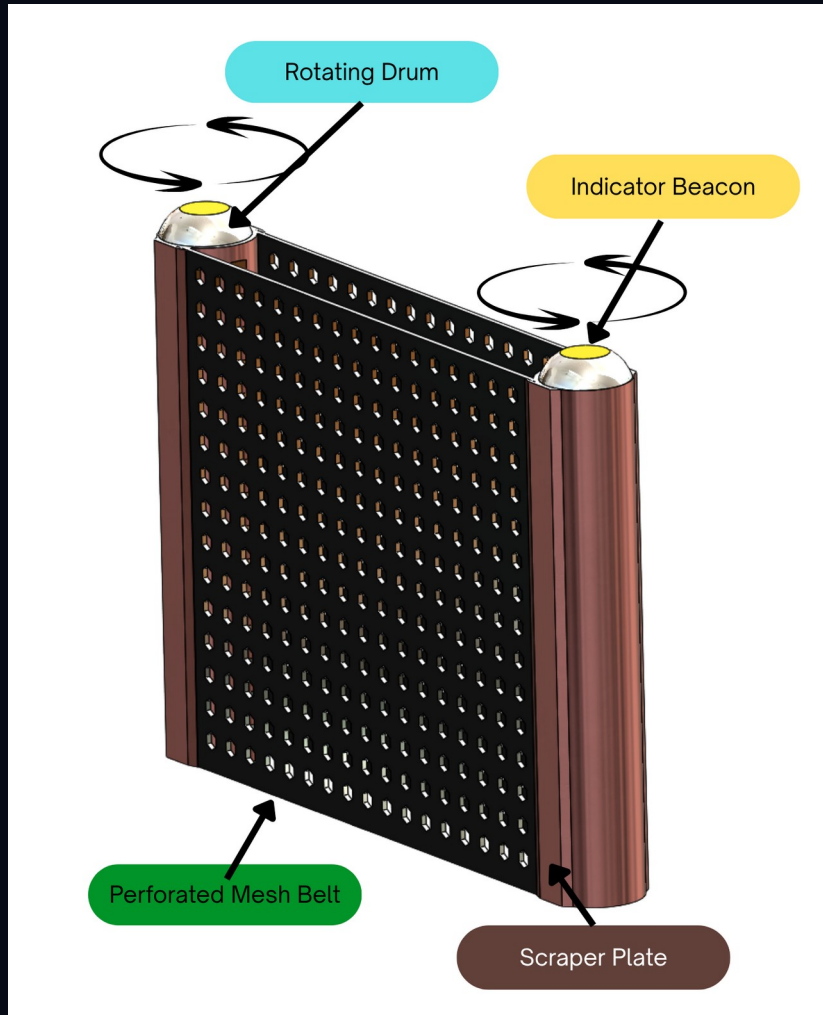
- Jellyfish reach an intake two ways: drifting in as adult medusae, or born locally as polyps on the channel's own walls.
- Layer 1 (mesh) stops adults entering from the Arabian Gulf at the inlet — the drift-in origin.
- Layer 2 (robot) stops polyps colonising breakwater, walls and forebay — the local-origin nursery.
- Either layer alone leaves the other origin open; together they close both pathways a bloom can use.
- Both pieces already exist separately — exclusion screens, AUVs, computer vision. Only the pairing for this channel is new.
- Fits ENEC's actual gap: management after the existing early-warning notice, with no new nuclear system.

Why Alternatives Fall Short

- Chlorination / chemical dosing — treats polyps but not adults already past the mouth; raises discharge and marine-life permitting concerns.
- Ultrasonic transducers — effectiveness is species- and site-dependent; covering a long channel's full wall area is impractical.
- Bubble curtain alone — lab and field trials show it is not adequate protection by itself; still needs a net or mesh behind it.
- Anti-fouling coating alone — stops polyps attaching, but does nothing to exclude adult jellyfish entering the mouth.
- Heat-shock treatment — kills on contact, but heating a channel this long is energy-intensive and disruptive to the Gulf's marine life.
- Diver / net clearance — slow, weather-limited and unsafe as the sole method inside a live nuclear-site intake.

How the rotating mesh avoids fouling

CAD concept for the inlet mesh module: a perforated belt loops continuously between two rotating pillars, like a traveling band screen. Each pass through the pillar brings the belt across a fixed scraper plate, so fouling never gets time to establish.



01 Continuous rotation

The belt loops between two pillars at all times. Every section cycles past the cleaning point on a short, fixed interval — no section sits still long enough for larvae to settle.

02 Scraper at each pillar

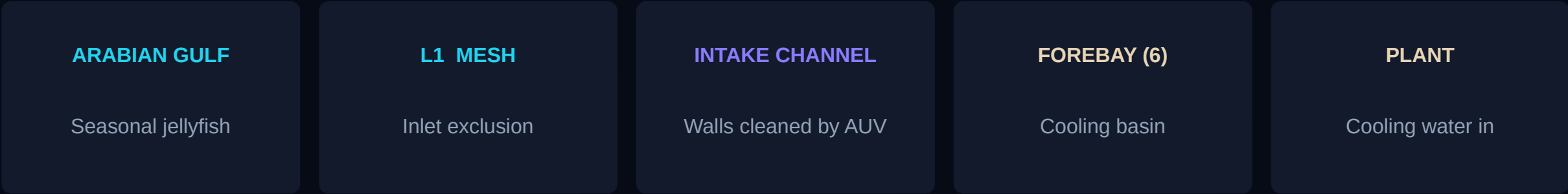
A fixed plate at the drive pillar mechanically scrapes the belt as it passes. Removes biofouling residue by disruption, not chemistry — no coating to reapply, no dosing to permit.

03 Modular, ganged sections

This CAD shows one section. Identical units are placed side-by-side to span the inlet, so a section can be pulled for maintenance without shutting the screen down.

One module shown. Multiple modules gang side-by-side to span the full inlet width.

Sea → Inlet mesh → Channel (robot) → Forebay → Plant



Barakah Nuclear Energy Plant — Facility Key

- | | |
|--|---|
| 1 Reactor Containment Buildings (Units 1-4) | 6 Cooling Water Intake Basin (Forebay) |
| 2 Turbine Halls | 7 Marine Breakwater |
| 3 Switchyard / Transformer Yard | 8 Intake / Outfall Channel (Discharge Turbulence) |
| 4 Support & Logistics Zone (Laydown Yards, Warehouses) | 9 Wharf (Heavy Component Offload Dock) |
| 5 Nawah Operations Building | 10 Airstrip |

Base imagery: satellite view of Barakah, Al Dhafra Region, Abu Dhabi, UAE

Mapped to Barakah

- 7 Marine breakwater
- Inlet Mesh spans the mouth
- Channel walls Robot patrol
- 6 Cooling-water intake basin
- 8 Intake / outfall channel

Robot loop: Detect → Remove → Collect → Continue

Hardware: AI camera, sonar, thrusters, arm + suction, LEDs, dock.

Protect the plant. Spare the sea. Spare the divers.

Cooling reliability

Fewer swarm events at the screens.
Intakes stay open when it matters.

Marine-life first

Exclusion, not mass killing. Gentle polyp removal. Coexistence with the Gulf.

Safer people

Cuts diver time in the channel and forebay after a bloom warning.

Source control

Stops incoming adults and the next generation growing on the walls.

ENEC alignment

Sits after existing detection. Post-warning management, not a new nuclear system.

Scalable pattern

Any coastal plant with a defined intake mouth and hard-walled channel can reuse the pathway.

Built from existing pieces, on a known geometry

Why it is credible

- Exclusion nets and travelling screens already protect NPP intakes worldwide.
- AUVs, computer vision and ROV arms are commercial technology.
- Barakah's inlet is a defined mouth — a mesh can span a known width.
- Robot task is repeatable: hard walls, mapped channel, after a warning.
- Modular: mesh and robot can be prototyped independently.
- Does not require redesign of plant or nuclear safety systems.

Assumptions and limits

- Mesh fouling is addressed by design: rotating belt + fixed scraper at each pillar (CAD validated). Spare modular sections still cover extended downtime.
- Vision drops in high turbidity; lighting and sonar fill the gap.
- Environmental and security approvals at a nuclear marine boundary.
- Does not replace travelling screens inside the plant.
- Models must be trained on local polyp morphology.
- Path: coupon tests → mock channel → supervised analogue pilot.

Prove the pathway — not nuclear operations

PHASE 1

Lab & tank

Mesh coupon: flow, jellyfish exclusion, self-clean cycle.
Vision accuracy on polyp models.
Arm force vs. concrete and rock.
Capture efficiency of suction.

PHASE 2

Mock channel

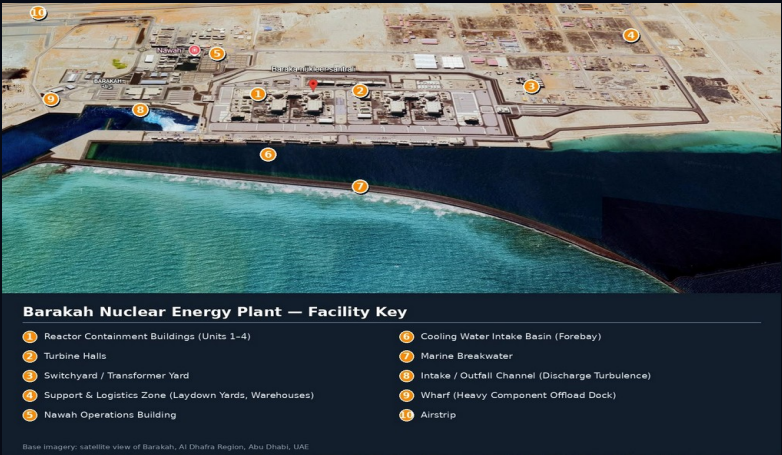
Scaled inlet + wall run.
Before / after polyp density.
Non-target organism checks.
Hours vs. a diver baseline.

PHASE 3

Site-like demo

Supervised analogue of Barakah intake geometry.
End-to-end: mesh hold + robot patrol.
Operational metrics for judges: coverage, success %, marine impact.

Site evidence + concept art



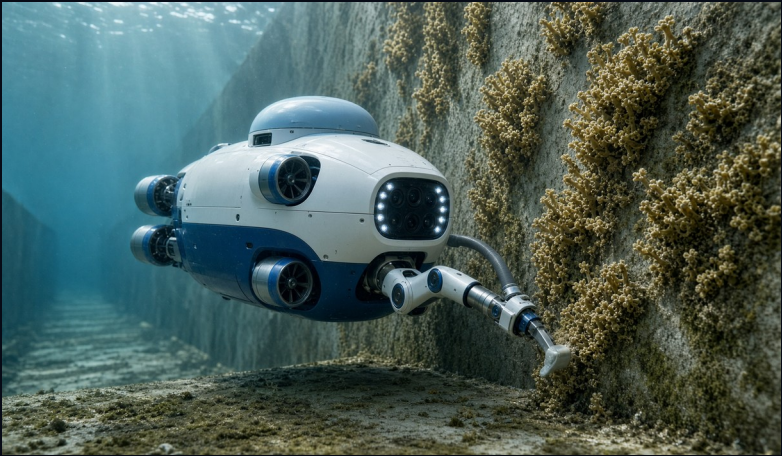
Facility key — 6 forebay, 7 breakwater, 8 channel



Inlet: proposed mesh location



Layer 1 — exclusion grid



Layer 2 — robot on walls



PolypClean hardware callouts



Both layers on the coastal intake

How might we advance sustainable bloom management?

After an early-warning, Barakah still has an open mouth to the sea and a long hard-walled channel. PolypClean answers with a dual pathway: a mesh that keeps swarms out, and a robot that stops polyps from turning the channel into a nursery. Manual intervention falls. Marine life is diverted, not destroyed. Plant cooling stays protected — without touching nuclear systems.

Site-specific

Designed on the real Barakah inlet, not a generic coastline.

Complementary

Sits after ENEC detection. Barrier + robotics, not either/or.

Sustainable

Keep out. Clean walls. Coexist with the Gulf.

Cleaner seas. A more balanced tomorrow.

Thank you · Team PolypClean · ENEC Student Challenge